

Job Seeker Copilot

A self-contained account of the working software, service architecture, verification evidence and intended AWS implementation.

Controlled private beta

31 repositories

28 application services

Confidence edition · v3

The screenshot displays the Job Seeker Copilot interface, which is divided into three main sections: Search, Applications, and Documents. The Search section is active, showing a list of job matches for 'Software Developer'. The results are sorted by 'Most relevant' and include details such as the job title, company, location, salary, and posting date. The Applications section shows a progress bar for 'Generated documents' at 100%, and the Documents section shows a progress bar for 'Applied' at 0%.

Search Applications Documents

REAL PROVIDERS Job search, matching and details

Job Matches Real providers Sort: Most relevant Filter Refresh

Software Developer (188) Software Engineer (Not searched) Full-Stack Developer (Not searched)

Java Developer (Not searched) Backend Developer (Not searched)

SHOWING 1-10 OF 188 MATCHES FOR SOFTWARE DEVELOPER · MORE AVAILABLE

Software Engineer REED.CO.UK DOCUMENTS PREPARED

Vermillion Analytics

London GBP 35,000 - 45,000 per year Posted Jun 8, 2026

Lead Full Stack Software Developer REED.CO.UK NOT SAVED

Spectrum IT Recruitment

Guildford GBP 80,000 - 100,000 per year Posted May 8, 2026

Junior Software Engineer REED.CO.UK DOCUMENTS PREPARED

Harnham - Data & Analytics Recruitment

London GBP 30,000 - 40,000 per year Posted Aug 7, 2026

Software Engineer (C/C+) REED.CO.UK DOCUMENTS PREPARED

Rise Technical Recruitment Limited

NOT ENABLED FOR THIS BETA

AI Credit & Payments

Wallet balance and purchasing will be enabled after the secure payment integration is approved.

Open AI Credit area

AVAILABLE Reporting & job-search evidence

Your job search progress Refresh

29 hours remaining this week.

Applications 6

Generated documents 6 100%

Applied 0 0%

Interviews 0 0%

Offers 0 0%

Unsuccessful 0 0%

Recent activity

11/08 20:36 Documents

Prepared for recruiters, engineering reviewers, potential partners and pilot organisations. The dossier explains the system without requiring access to private source repositories.

Start with confidence; continue into the engineering

The opening pages establish what is working and how it has been verified. The middle explains product behaviour, data ownership and operational boundaries. The appendix describes every repository and its runtime communications so a reader does not need private GitHub access.

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Evidence boundary. Present-day claims were checked against source, contracts and the running environment. Historical claims use Git/PR/issue evidence. Dated reports prove only their named conditions.

One workspace for the fragmented work of finding a job

Job Seeker Copilot helps a candidate organise career evidence, discover vacancies, prepare tailored documents, track applications and retain an auditable record of what was used.

The product is a connected workflow: structured evidence and a genuine vacancy become versioned, application-specific documents and an owned application record.

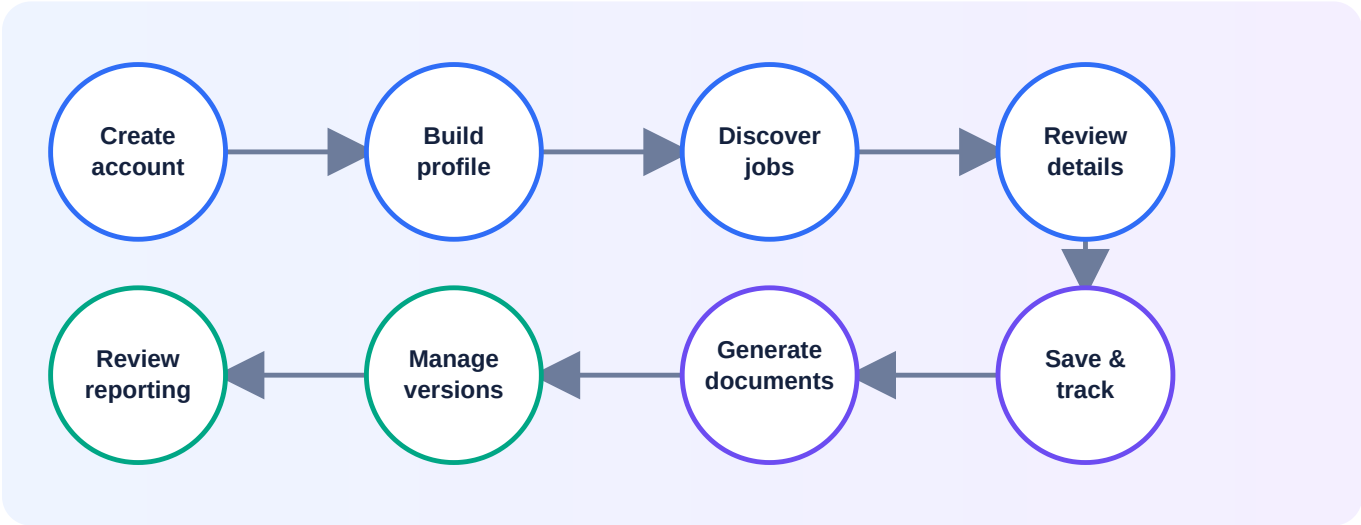
Candidate problem
Vacancies, versions, portals, evidence and follow-up are scattered across tools.

Product response
Structured profile, multi-provider discovery, document lineage and application tracking.

Engineering response
Owned domain state, provider gateways, versioned contracts and recoverable workflows.

Current boundary
Controlled private beta; local persistent operation, not public production.

31	28	7	5
repositories audited—not used as a vanity metric	application services in the audited manual profile	independent PostgreSQL state owners	job-provider domains



Confidence comes from several independent kinds of evidence

28

application services passed the audited persistent-manual health contract

32/32

retained full Cucumber regression scenarios passed on 11 August

25/25

fixture-backed concurrent discovery journeys completed in the measured ladder

PASS

live-provider/OpenAI showcase scenario completed on 12 August

Working product paths

- ✓ registration, sessions and persistent profiles;
- ✓ five-source vacancy discovery and detail;
- ✓ save, document preparation and tracking;
- ✓ reporting and exact version history.

Durable engineering

- ✓ seven PostgreSQL state owners;
- ✓ immutable evidence/job/document references;
- ✓ idempotent multi-service operations;
- ✓ producer-owned OpenAPI contracts.

Real integration evidence

- ✓ live provider-backed search;
- ✓ real Google location path;
- ✓ real OpenAI document generation;
- ✓ persistent restart-safe manual account.

Repeatable verification

- ✓ System Data canonical personas;
- ✓ Playwright and Cucumber journeys;
- ✓ component, contract and policy gates;
- ✓ executable video recording.

Measured—not assumed

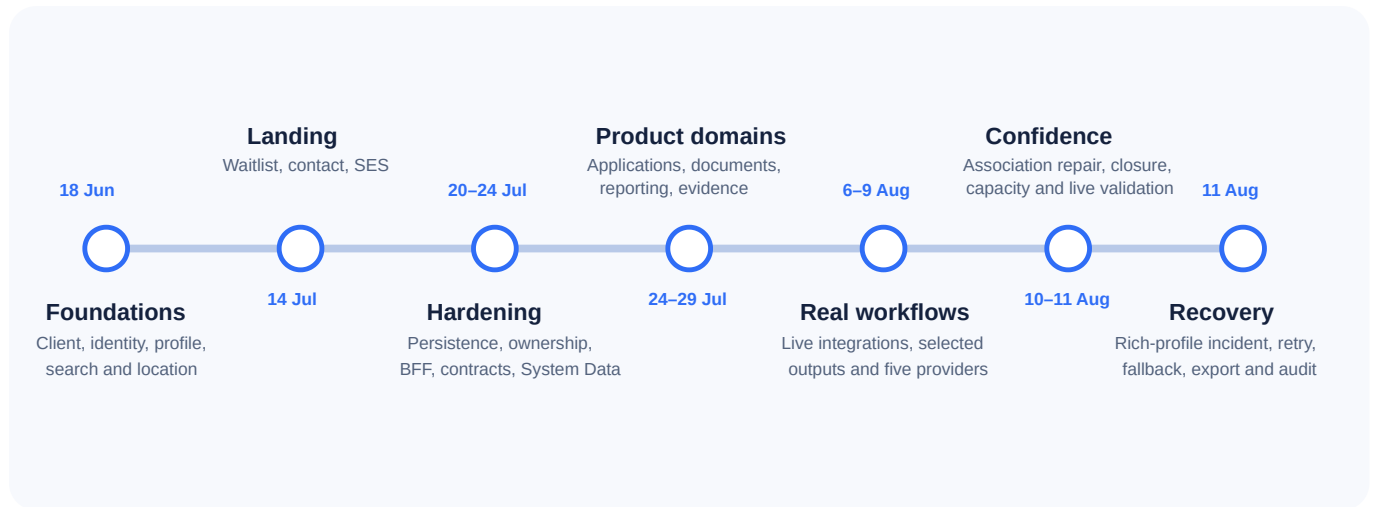
- ✓ 1/5/10/15/20/25-session ladder;
- ✓ latency, CPU and memory retained;
- ✓ zero restarts/OOMs in final ladder;
- ✓ 25 is functional, not comfortable.

Current boundary

- ✓ controlled beta, not public production;
- ✓ AWS shape is calculated, not tested;
- ✓ live Stripe checkout remains disabled;
- ✓ security/HA assurance remains gated.

What this proves. The application is not a staged interface: browser actions cross authenticated gateways, domain services, owned databases and external-provider boundaries, and the resulting state survives ordinary service restarts.

A concentrated sequence of architectural maturation



- 18 June** Initial client, authentication, profile, location, job search and provider repositories establish the application foundation.
- 14–22 July** Public landing/waitlist work is followed by identity hardening, durable migrations, HttpOnly session architecture, vulnerability gates and accessible flows.
- 22–24 July** System Data and Playwright/Cucumber are imported; generated client binaries are removed; producer-owned contracts and clean builds become policy.
- 24–29 July** Application tracking, document storage/export, reporting, payments, evidence revisions, snapshots and claim provenance become separate owned domains.
- 6–9 August** Real OpenAI paths, retained generation, exact document selection, uploads, Location Service/Google isolation and five-source discovery are integrated.
- 10–11 August** Realistic E2E exposes application/document association gaps; private-beta closure, live providers and capacity validation follow.
- 11 August** A rich real profile exposes duplicate structure and output/deadline weaknesses. Repair, bounded retry, unknown-outcome reconciliation, deterministic fallback and export quality are strengthened.

History boundary. The repositories visible here were migrated or created in June–August 2026; this timeline describes evidenced repository history, not undocumented work before the recorded baselines.

Every screen is backed by an ownership decision



Jobseeker Copilot

Create your account, then shape a job search that works for you.

Create accountSign in

Your account details

No CV, work history, skills or postcode required.

Full name

Use between 1 and 100 characters.

Email address

Password

Use 15 to 128 characters. Spaces are kept exactly.

Create account

[Job Seeker Copilot](#) [Privacy Policy](#) [Terms of Use](#)

Registration. Minimal first entry; no CV or postcode required. Browser tokens remain in HttpOnly cookies.

Software Developer (188)Software Engineer (Not searched)Full-Stack Developer (Not searched)

Java Developer (Not searched)Backend Developer (Not searched)

HOWING 1-10 OF 188 MATCHES FOR SOFTWARE DEVELOPER · MORE AVAILABLE

Software Engineer

Vermillion Analytics

LondonGBP 35,000 - 45,000 per yearPosted Jun 8, 2026

REED.CO.UKDOCUMENTS PREPARED

Lead Full Stack Software Developer

Spectrum IT Recruitment

GuildfordGBP 80,000 - 100,000 per yearPosted May 8, 2026

REED.CO.UKNOT SAVED

Junior Software Engineer

Harnham - Data & Analytics Recruitment

LondonGBP 30,000 - 40,000 per yearPosted Aug 7, 2026

REED.CO.UKDOCUMENTS PREPARED

Software Engineer (C/C+)

Rise Technical Recruitment Limited

StevenageSalary not specifiedPosted date unavailable

REED.CO.UKDOCUMENTS PREPARED

Software Developer Trainee

ITOL Recruit

E62RPGBP 28,000 - 35,000 per yearPosted date unavailable

REED.CO.UKNOT SAVED

Lead AI Software Engineer

Avanti

W60QUGBP 80,000 - 120,000 per yearPosted date unavailable

REED.CO.UKNOT SAVED

Discovery. Provider status, source attribution and application state are visible rather than hidden.



User step	Backend engineering	Durable truth
Register/sign in	BFF cookie and CSRF boundary; UMG; RS256/JWKS	Authentication DB
Build profile	Progressive sections, revisions, confirmation and provenance	Profile DB
Discover/save	Five-provider fan-out, mapping, dedupe and snapshot	Job DB
Generate	Evidence snapshot, wallet reservation, LLM validation and fallback	Generation ledger + Store
Apply/track	Exact document selections, immutable events and transitions	Application DB
Report	Read-only projections from owner-scoped services	Upstream owners

Decomposed by ownership, corrected by failure

One authoritative owner

Accounts, profile evidence, saved vacancies, applications, documents, wallet entries and operation checkpoints have explicit systems of record.

Gateways isolate trust

Browser, provider and model DTOs/credentials stop at controlled boundaries. Domain services consume canonical contracts.

Immutable references

Evidence snapshots, saved jobs, document versions and applied selections resist later mutation.

Failure is state

Retries, ambiguous outcomes and partial side effects are persisted and reconciled rather than hidden behind a generic error.

Evolution pattern

Original simplification

Generate content, save output and continue synchronously.



Current operation

Persist intent, snapshot inputs, reserve credit, checkpoint side effects, validate output, approve exact versions and reconcile application references.

Why microservices?

Different trust, state and failure boundaries—not repository count for its own sake.

Why contracts?

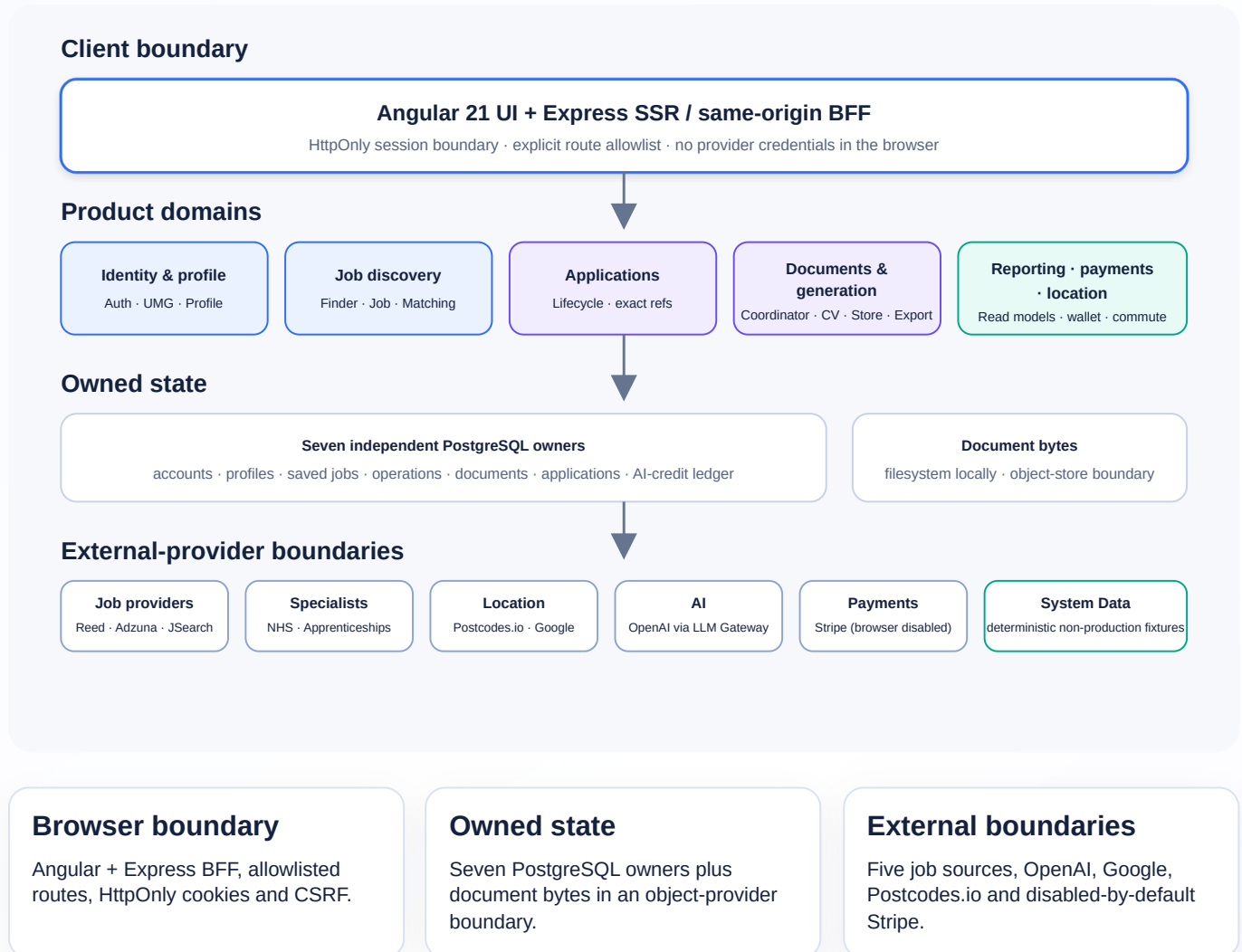
Independent repositories otherwise compile against yesterday's assumptions.

Why fixtures?

Repeatable failure and zero-spend regression, kept distinct from live confidence.

Architecture was not fixed in advance. Document association, rich evidence, provider inconsistency and realistic concurrency each forced clearer ownership or recovery rules.

Major boundaries without a 30-service spaghetti diagram



Arrows show material runtime calls, not every health, lifecycle or fixture-control edge. The full catalogue appears on pages 39–45.

Six principal call paths explain the service topology

Identity and profile

Browser → Client BFF → User Management Gateway

UMG calls Authentication for sessions and User Profile for profile/evidence. Downstream services validate the authenticated subject independently.

Vacancy discovery

Client → Job Finder → Job Service

Job Service fans out to Reed, Adzuna, JSearch, NHS and Apprenticeships gateways, then calls Matching for application and commute enrichment.

Application documents

Client → Document Generation Gateway

The coordinator calls Profile, Job, CV/Letter, LLM, Payment, Store, Export and Application Tracker through durable checkpoints.

Applications and reporting

Client → Job Finder / Reporting Gateway

Application Tracker owns lifecycle truth. Reporting Service reads Tracker, Store and Profile without mutating any owner.

Location and commute

Client → Location Gateway → Location Service

Location Service calls Google Maps and Postcodes.io gateways; Job Matching consumes only provider-neutral location/route results.

Deterministic verification

E2E → Client + System Data

Fixture-mode gateways read governed provider/persona data. Live profiles replace only approved external boundaries while keeping application ownership unchanged.

31

reviewed repositories

27

Java/Spring applications

7

PostgreSQL owners

28

composed application processes

Reading rule. An arrow means an authenticated/runtime API call, not shared-database access. Pages 39–45 name every repository, its state and both sides of its communication boundary.

The Angular application is a product boundary, not a thin skin

Major user areas

- ✓ Registration, sign-in, refresh and return
- ✓ Dashboard, profile, preferences and evidence manager
- ✓ Search, provider status, job details and commute
- ✓ Evidence selection and asynchronous generation
- ✓ Documents, versions, archive/recovery and downloads
- ✓ Applications, exact selections, status and reporting

Engineering behaviour

- ✓ Generated/typed API models where contracts exist
- ✓ Same-origin Express route allowlist
- ✓ HttpOnly session cookies; no bearer token in localStorage
- ✓ CSRF on state-changing proxy routes
- ✓ Provider URLs/content treated as untrusted
- ✓ Loading, partial, empty, expired and recoverable states

```
app.post('/api/auth/refresh', async (req, res) => {
  await proxyUserManagementRequest('/api/auth/refresh', 'POST', req, res, true);
});
if (forwardSessionCookies && result.setCookies.length) {
  res.setHeader('Set-Cookie', result.setCookies);
}
registerDocumentGenerationRoutes(app, {
  accessCookieName, csrfCookieName, origin, timeoutMs
});
```

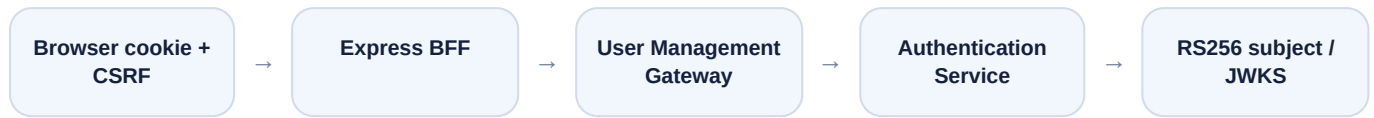
Why the BFF exists. Browser code never chooses arbitrary service destinations or stores access tokens. The server boundary converts cookies into controlled downstream requests and stable, redacted failures.

The dashboard interface is divided into several sections:

- Top Navigation:** Search, Applications, Documents.
- REAL PROVIDERS:** Job search, matching and details.
- Job Matches:** Real providers, Sort: Most relevant, Filter, Refresh.
- Filters:** Software Developer (188), Software Engineer (Not searched), Full-Stack Developer (Not searched), Java Developer (Not searched), Backend Developer (Not searched).
- Showing 1-10 of 188 matches for SOFTWARE DEVELOPER · MORE AVAILABLE**
- Job Listings:**
 - Software Engineer** (Vermillion Analytics, London, GBP 35,000 - 45,000 per year, Posted Jun 8, 2026). Status: REED.CO.UK, DOCUMENTS PREPARED.
 - Lead Full Stack Software Developer** (Spectrum IT Recruitment, Guildford, GBP 80,000 - 100,000 per year, Posted May 8, 2026). Status: REED.CO.UK, NOT SAVED.
- NOT ENABLED FOR THIS BETA:** AI Credit & Payments. Wallet balance and purchasing will be enabled after the secure payment integration is approved. [Open AI Credit area](#)
- AVAILABLE:** Reporting & job-search evidence.
- Your job search progress:** 29 hours remaining this week. Refresh.
- Applications:** 6
- Progress Tracking:**
 - Generated documents: 6 (100%)
 - Applied: 0 (0%)
 - Interviews: 0 (0%)

The dashboard composes application, document and reporting state from separate owners without becoming their database.

Identity is derived, forwarded and independently enforced



Authentication owns

- canonical email identity and password state;
- registration, login, refresh, logout and reset;
- refresh/session lifecycle in PostgreSQL;
- asymmetric token signing and public JWKS.

Consumers enforce

- validated token subject, never caller-supplied user ID;
- route-scoped service identities;
- owner-scoped reads and writes;
- redacted browser-safe error contracts.

Architecture change

Earlier risk

Bearer/session material in browser storage and caller-controlled downstream identity widened the impersonation boundary.



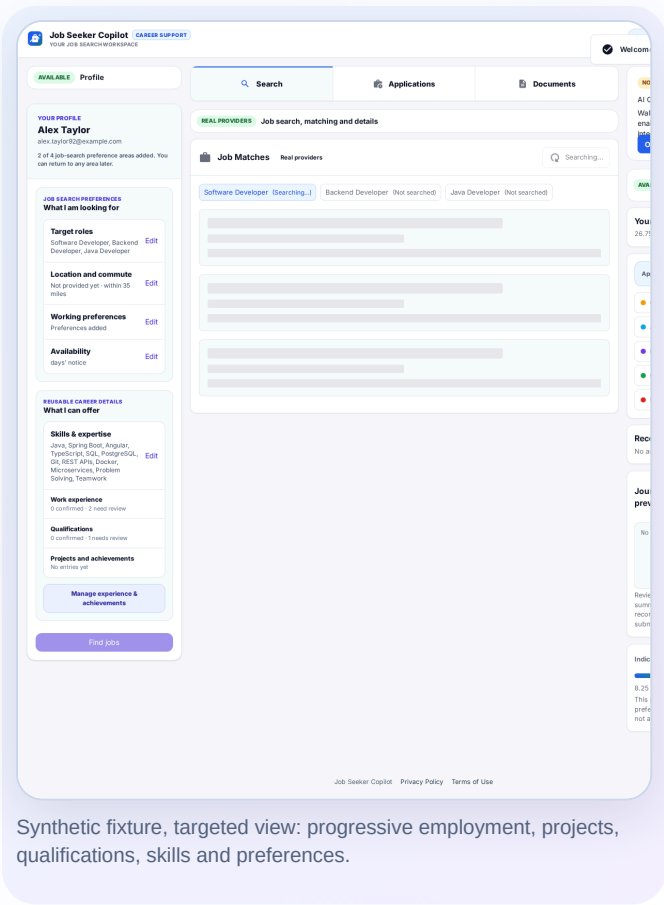
Current boundary

HttpOnly cookies terminate at the same-origin BFF; gateways/services derive and verify the subject independently.

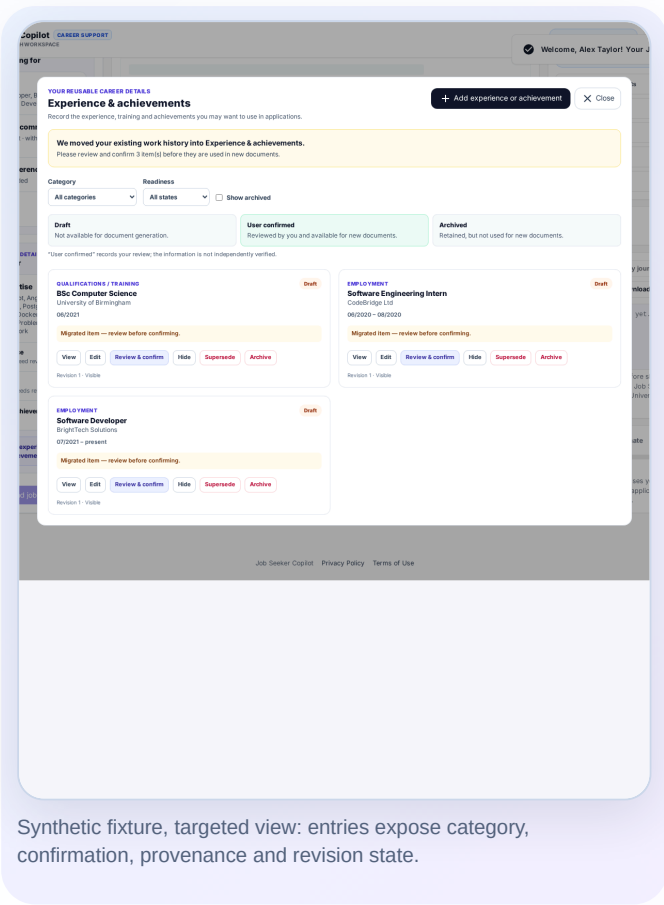
Control	Evidence	Limit
Credential persistence	Migration-backed PostgreSQL	Local/private-beta operation is not a security certification.
Abuse resistance	Bounded login attempts, input normalisation, stable errors	Production threat monitoring still requires deployed operations.
Lifecycle	Refresh/logout/reset and account cleanup calls	Cross-service retention policy remains ongoing product work.

Evidence basis: authentication/session contracts, BFF route policy, database migrations, JWKS verification and retained registration/return journeys.

Career history is structured product data



Synthetic fixture, targeted view: progressive employment, projects, qualifications, skills and preferences.



Synthetic fixture, targeted view: entries expose category, confirmation, provenance and revision state.

Entry

Stable category and identity; visibility, lifecycle and review state.

Revision

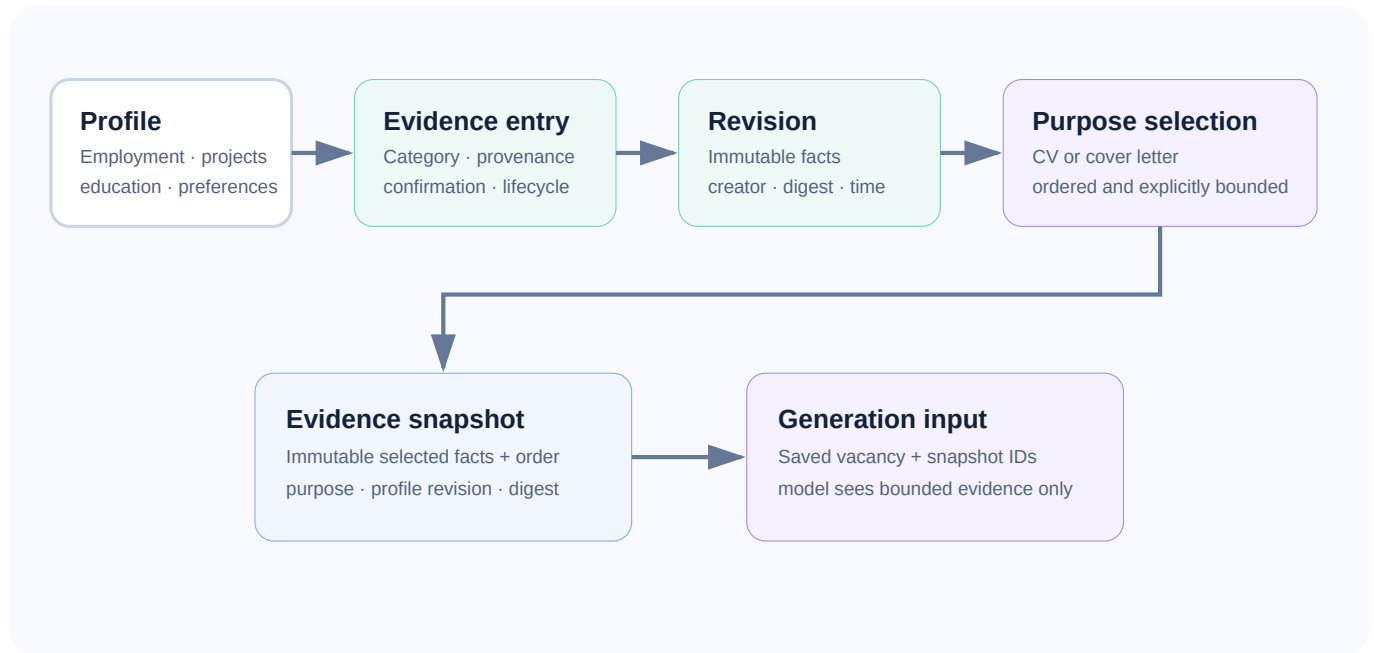
Immutable facts, creator, confirmation and content digest.

Snapshot

Purpose, selected revisions, section order, profile revision and digest.

Why this exists. Repeatedly pasting a free-text biography into a model makes provenance and later correction impossible. Structured evidence allows a user to confirm facts once, select them for a purpose and retain exactly what supported a generated document.

Selection becomes an immutable generation input



```

EvidenceSnapshot create(userId, request) {
  require(profile.revisionId && profile.contentDigest);
  for (entryId : request.entryIds) {
    revision = eligibleUserConfirmedRevision(entryId, userId);
    enforceSectionOrderAndFactLimit(revision);
    snapshot.addSelection(copyFactsAndDigests(revision));
  }
  snapshot.snapshotDigest = digest(snapshot);
  return repository.saveAndFlush(snapshot);
}
  
```

CV purpose

Allows broad but bounded career evidence and selected section order.

Cover-letter purpose

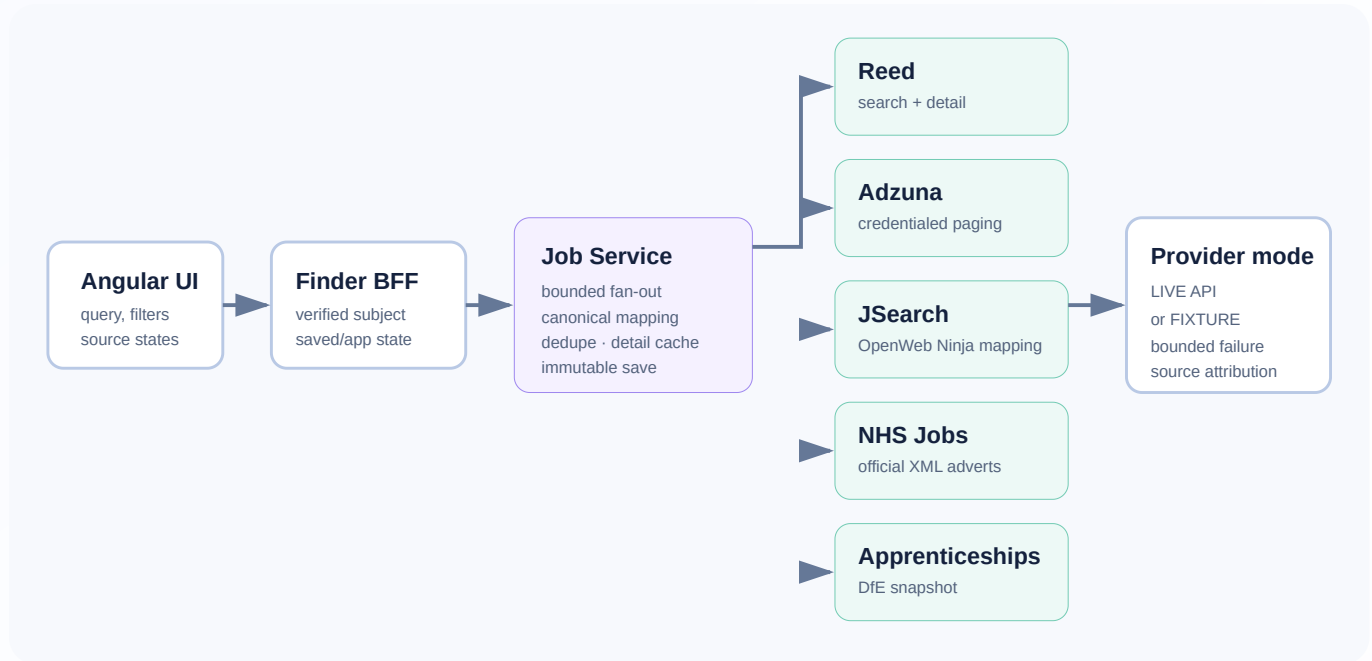
Can use a narrower, vacancy-specific evidence subset.

Application provenance

Snapshot and claim-ledger IDs/digests travel with document versions.

Boundaries. Duplicate selections, unconfirmed/latest-invalid revisions, hidden/review-required evidence and oversized snapshots fail before the model call.

Five providers become one truthful product response



Fan-out

Provider calls are bounded independently. One failure can produce a partial response rather than erase healthy sources.

Mapping

Adapters convert provider IDs, dates, salary, workplace type, descriptions and source URLs.

Save

The selected result is hydrated where supported and persisted as an immutable vacancy snapshot.

Software Developer (188) Software Engineer (Not searched) Full-Stack Developer (Not searched)

Java Developer (Not searched) Backend Developer (Not searched)

SHOWING 1-10 OF 188 MATCHES FOR SOFTWARE DEVELOPER · MORE AVAILABLE

Software Engineer REED.CO.UK DOCUMENTS PREPARED ▾

Vermillion Analytics

📍 London 💰 GBP 35,000 - 45,000 per year 📅 Posted Jun 8, 2026

Targeted point-in-time live-provider view. Changing external adverts and provider uptime are not regression assertions.

Each source has different contracts and failure shapes

Provider	Authentication/acquisition	Mapping/detail behaviour	Modes and known limits
Reed	API key; bounded search/detail calls	Dedicated detail endpoint, source/apply URLs, salary and posted data	Fixture default; bounded live sample; advert text can be incomplete
Adzuna	App ID/key; paged search	Credentialed pagination, workplace type and canonical mapping	Fixture default; paging/quota and provider date differences
JSearch	OpenWeb Ninja API key	Provider records mapped from aggregated upstream sources	Fixture default; IDs, descriptions and dates vary
NHS Jobs	Official self-serve XML path	Authoritative NHS identity and advert/source provenance	Live path plus deterministic in-service fixture
Apprenticeships	DfE Display Advert API v2 credentials	Authenticated acquisition to atomic snapshot; training details	Readers see last complete snapshot, never half-refresh state

SearchApplicationsDocuments

REAL PROVIDERSJob search, matching and details

Job MatchesReal providers — partial availabilitySortMost relevantFilterRefresh

Software Engineer (183)Software Developer (191)

Job Seeker CopilotCAREER SUPPORTYOUR JOB SEARCH WORKSPACE

AVAILABLEProfile

YOUR PROFILE

Alex Tayloralex.taylor02@example.com3 of 4 job-search preference areas added. You can return to any area later.

JOB SEARCH PREFERENCES

What I am looking for

Target rolesSoftware Engineer, Software DeveloperEdit

Location and commuteReading, South East (RG1 1AA) - within 35 milesEdit

Working preferencesPreferences addedEdit

Availabilitydays' noticeEdit

REUSABLE CAREER DETAILS

What I can offer

SHOWING 1-10 OF 191 MATCHES FOR SOFTWARE DEVELOPER · MORE AVAILABLE

Software Engineer (AWS, Python, C#/C++, CFD/FEA)

REED.CO.UKNOT SAVED

Hays Specialist Recruitment Limited

OxfordGBP 45,000 - 68,000 per yearPosted date unavailable

Job Description

Software Engineer (AWS, Python, C#/C++, CFD/FEA)
Oxford - 3+ days per week in office.
£45000 - £68000 + Package.

- Must have a Computing/STEM Degree (2:1 or higher).

- Can work in their Oxford head office 3+ days week.

- Must have experience with AWS / Distributed Computing and Applied/Mechanical Engineering

Your new company

You'll be joining a cutting-edge engineering and technology organisation that has grown rapidly in the UK, now pushing into global markets. Formed by world leading experts in science/software the business is redefining how high performance components/instruments are designed.

This role would essentially suit someone with a passion for Mechanical Engineering/Physical Modelling with commercial Software Development experience - you must have written production grade code in C#, C++ or Python for application development (not web apps or embedded) and an interest in pushing Machine Learning over time.

Crucially we are looking for experience with AWS Cloud / Distributed Computing to be considered.

Your new role

In this role you'll be tackling engineering challenges that most organisations simply can't touch and building proprietary software that makes a difference in the world. You will thrive in an environment where new challenges

NOT ENABLED FOR THIS BETA

AI Credit & Payments

Wallet balance and purchasing will be enabled after the secure payment integration is approved.

Open AI Credit area

AVAILABLEReporting & job-search evidence

Your job search progress

26.75 hours remaining this week.Refresh

Applications9

Generated documents333%

Applied222%

Interviews111%

Offers111%

Unsuccessful111%

Targeted view of a synthetic expanded advert, provider status, source and generation controls. Provider text is displayed as untrusted content.

Engineering work beyond “calling an API”. Authentication, paging, XML/JSON shape, missing fields, status/date semantics, source identity, provider errors, attribution, terms-safe caching and deterministic equivalents all differ.

Job-provider boundaries

15

Heterogeneous adverts need explainable identity

Canonical fields retained

- ✓ canonical job ID and provider/external IDs;
- ✓ all source references and canonicalised URLs;
- ✓ title, company, location and workplace type;
- ✓ salary, posted/expiry dates and description;
- ✓ specialist/apprenticeship details and provenance.

Duplicate signals

- ✓ same canonical apply URL;
- ✓ overlapping source/apply options;
- ✓ normalised company + high title similarity;
- ✓ normalised location + compatible dates;
- ✓ compatible descriptions within configured thresholds.

```
if (sameCanonicalApplyUrl(left, right) || overlappingApplyOptions(left, right)) return 1.0;
if (sameCompany(left, right)
    && titleSimilarity(left, right) >= 0.90
    && sameLocation(left, right)
    && postedDatesCompatible(left, right)
    && descriptionsCompatible(left, right)) return 1.0;

mergeDuplicateJobs(primary, duplicate); // also merges every source reference
```

Naïve result

Two apparently identical cards or, worse, one merged record that loses the source needed to apply.



Current result

One canonical display record can retain multiple provider references, a preferred apply URL and field provenance.

Known boundary. Deduplication is deliberately rule-based and explainable. It can still miss semantically identical adverts or avoid merging uncertain ones; it is not an opaque similarity model.

A saved vacancy becomes owned lifecycle state

Software Engineer

Harnham - Data & Analytics Recruitment • London

Created 11 Aug, 20:34 •

DOCUMENTS PREPARED

CV and cover letter saved

CV Generated
11 Aug, 20:34

Cover Letter Generated
11 Aug, 20:34

Documents Uploaded

Mark Applied

Interview

Offer

Documents for this application

Choose exact approved versions; Recommended does not save automatically.

Experience & achievements used

Selected for the current document drafts

REED

View Application

Targeted view: application state, milestones and exact document selection remain visible after generation.

Identity

Owner + canonical/provider/external job identity; idempotent creation independent of generation.

Lifecycle

SAVED →
DOCUMENTS_GENERATED →
APPLIED → INTERVIEW → OFFER
→ ACCEPTED, with guarded transitions.

History

Immutable events and milestone dates survive refresh and reporting projection.

```
PUT /applications/{id}/document-selections
// CV and cover-letter slots are both mandatory:
// SELECTED(exact approved version) or OMITTED.
// Store verifies owner, type, availability and provenance.
// APPLIED freezes both choices atomically.
```

Separation of truth. Document Store proves what a version is and whether it is available. Application Tracker decides which exact versions the candidate selected and whether that choice is now frozen.

Evidence basis: Tracker and Generation contracts, immutable-event persistence, Store reference verification and the retained application/document-selection journey.

A document is a family, immutable versions and files —not one mutable blob

Document families

23

Each family is counted once

CV families

17

All retained versions

Cover-letter families

6

All retained versions

Document families 23

CVs 17

Filters 6

Search document families

Sort by Recently up

Document families

Cover letter — Software Engineer

Cover letter

1 retained version Latest v1

AI-tailored

Job scope

Software Engineer — Harnham - Data & Analytics Recruitment

Open for associations

No current version

CV — Software Engineer

CV

1 retained version Latest v1

AI-tailored

Job scope

Software Engineer — Harnham - Data & Analytics Recruitment

Open for associations

No current version

CV — Software Developer

CV

1 retained version Latest v1

AI-tailored

Job scope

Software Developer — Reed

Open for associations

No current version

CV — Software Developer

CV

1 retained version Latest v1

AI-tailored

Job scope

Software Developer — Reed

Open for associations

No current version

CV — Software Developer

CV

1 retained version Latest v1

AI-tailored

Job scope

Software Developer — Reed

Open for associations

No current version

Targeted view: families expose type, job scope, source, retention, current state and application associations.

Document Store owns

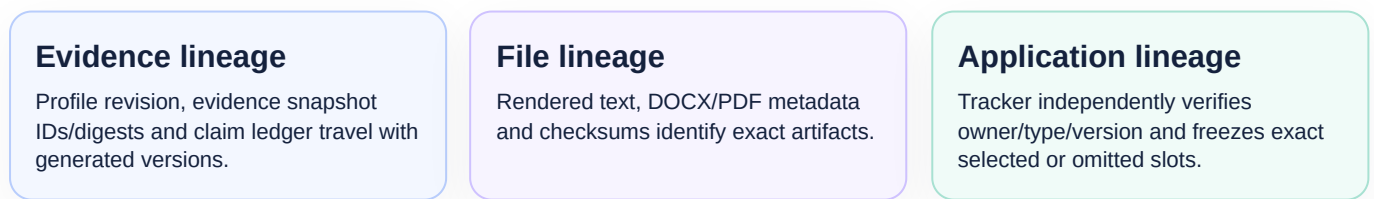
- ✓ family and immutable version records;
- ✓ generated/uploaded source and lineage;
- ✓ current/approved pointers;
- ✓ DOCX/PDF metadata, checksum and MIME;
- ✓ filesystem bytes locally or S3-compatible provider;
- ✓ archive, restore, recoverable delete and tombstones.

Upload boundary

PDF/DOCX ≤10 MiB; bounded extraction; MIME/signature checks; isolated ClamAV scanning; no model or AI-credit call.

Lifecycle action	Meaning	What it does not do
Set current	Select family recommendation	Does not rewrite an application
Approve	Make exact version eligible	Does not auto-select it for a job
Archive/restore	Reversible organisation state	Does not mutate bytes
Delete/recover	Recovery-window lifecycle	Purge remains separately controlled

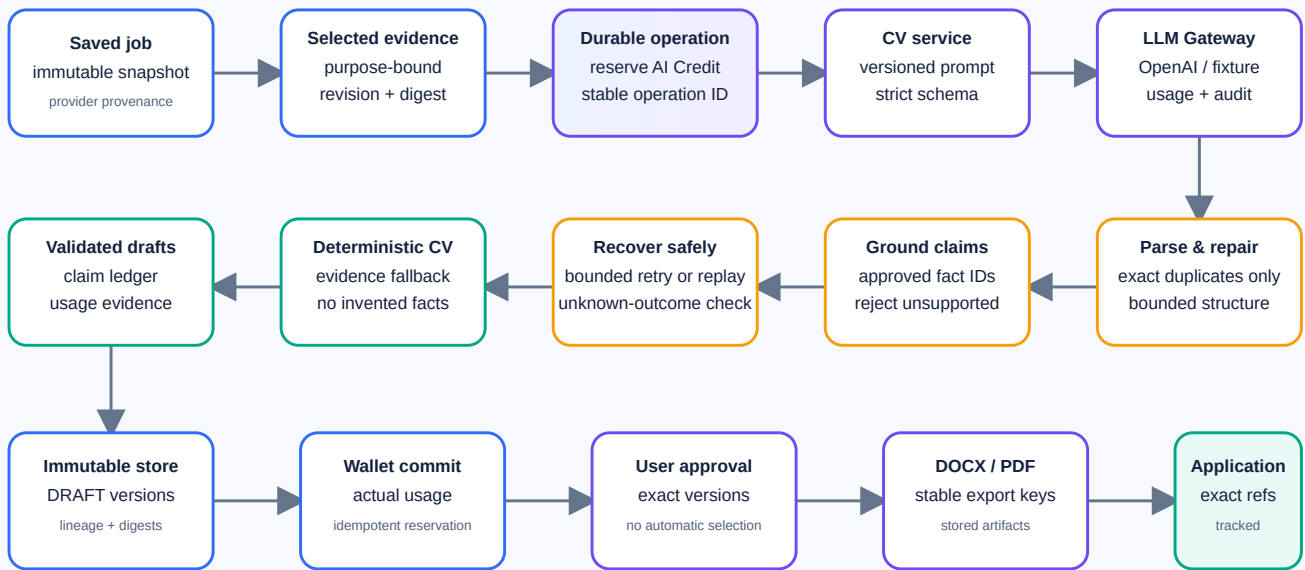
Historical problem solved. Regenerating a CV tomorrow must not silently change the version associated with yesterday's application.



Version and application truth

Generation is a durable product operation

The LLM is a bounded collaborator, not a source of truth



Before the model

Persist operation/deadline; resolve saved vacancy; create purpose snapshots; estimate and reserve credit.

After the model

Validate claims; store draft versions; commit actual usage; wait for explicit approval.

After approval

Set pointers; export exact versions; create/reconcile application and exact document selections.

Software Engineer

REED.CO.UK DOCUMENTS PREPARED

Vermillion Analytics

London

GBP 35,000 - 45,000 per year

Posted Jun 8, 2026

Job Description

Software Engineer Location: Hybrid / Remote (UK-based) Team: Engineering Salary: Up to 45000 Reports to: VP of Engineering You must be UK based and eligible to work. About Our Client Our client is one of Europe's fastest-growing software companies, on a mission to solve productivity problems for people-intensive industries using AI and automation. Since being founded, they've focused on automating the manual, tedious parts of complex workflows...

Read full advert

View on Reed.co.uk

CV and Cover Letter generated successfully

CV

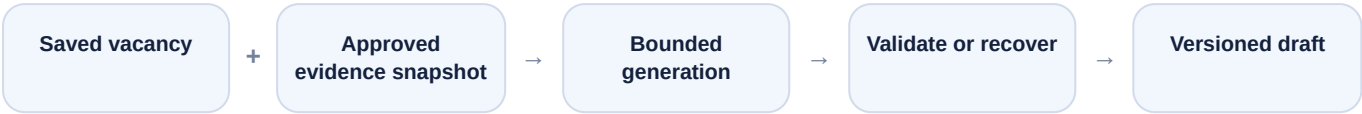
Download DOCX
Download PDF
Upload CV

COVER LETTER

Download DOCX
Download PDF
Upload Letter

Real vacancy and provider path; exact generated drafts expose DOCX/PDF only after the operation completes.

Generated content is useful only after the product can trust it



Bounded request

The service sends the vacancy, ordered evidence, purpose and schema version—not unrestricted profile data. Trusted contact identity stays outside generated content.

Typed validation

Strict parsing, structural limits and evidence-path checks reject unsupported qualifications, dates, metrics, employers or technologies.

Safe recovery

Exact duplicate structure may be repaired and revalidated. Classified retry is bounded; uncertain provider outcomes are reconciled before any new paid call.

Condition	Product response	Protection
Harmless exact duplicate	Remove duplicate, then repeat full validation	No semantic rewriting
Unsupported claim	Reject generated draft	Approved evidence remains authoritative
Known retryable provider failure	One bounded retry under the same operation	Stable operation and wallet keys
Outcome unknown	Reconcile retained outcome before deciding	No blind duplicate call or charge
No acceptable generated CV	Create a conservative evidence-based fallback	Copied facts only; provenance retained

Failure exposed by realistic use

A rich profile produced duplicate structure and then reached the previous output/deadline boundary—conditions compact fixtures had not represented.

→

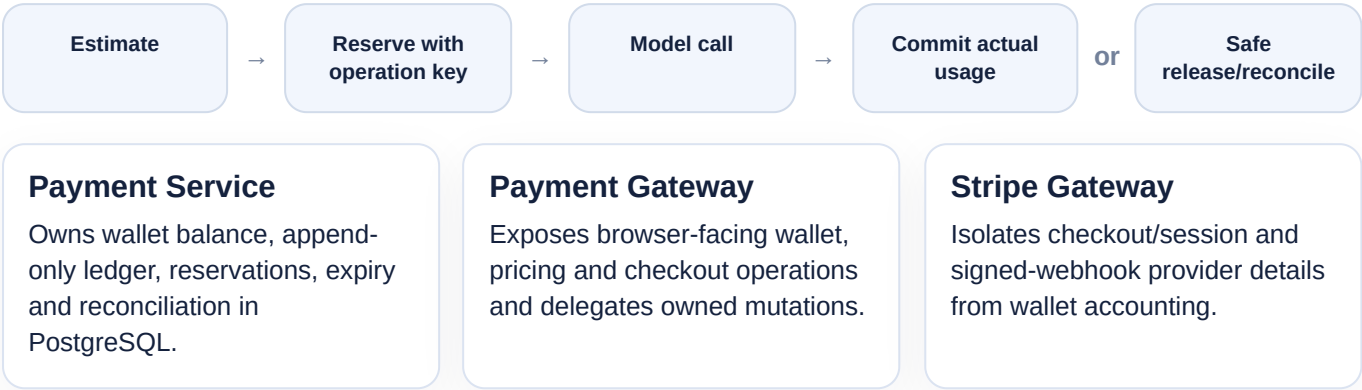
General product correction

Budgets, repair, claim validation, selected-output recovery, outcome reconciliation and deterministic fallback now form one tested recovery policy.

Why this matters. The product can reject an unacceptable generated response without losing the user’s evidence selection, duplicating a wallet charge or pretending the response was valid.

Repeating an export operation uses the exact source version and stable key; it does not silently create unrelated lineage.

Paid operations need accounting behaviour, not just a checkout button



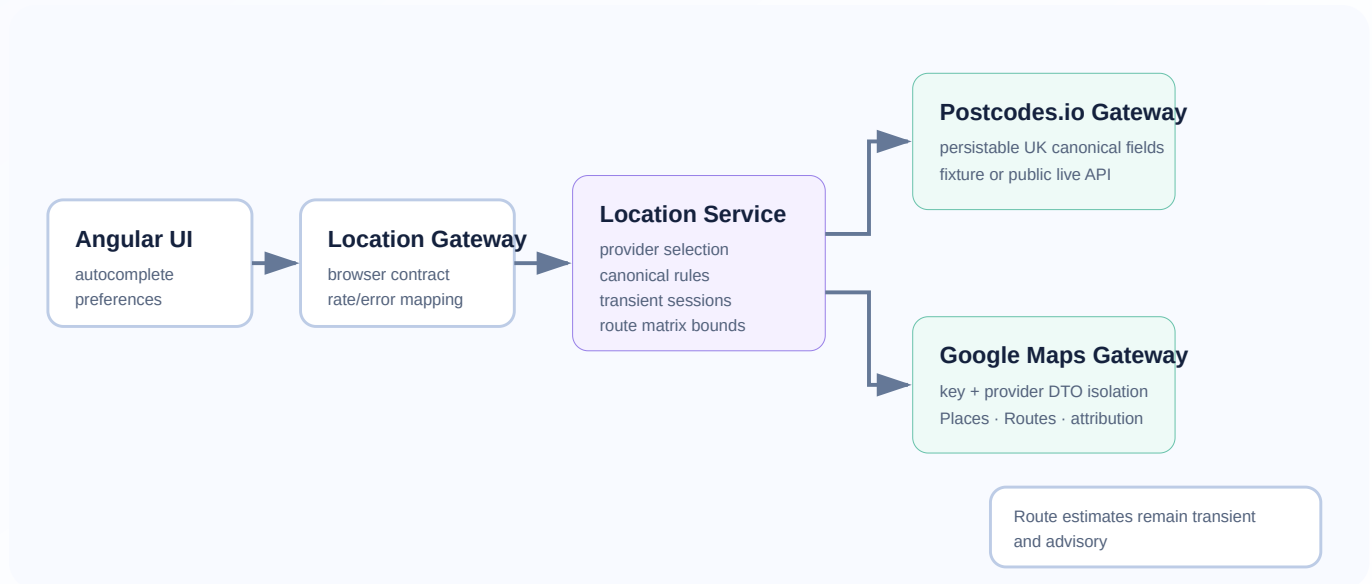
Reservation state	Allowed meaning	Retry property
RESERVED	Credit temporarily unavailable while operation is active	Same operation key returns/reconciles same reservation
COMMITTED	Actual model usage charged	Repeated commit is idempotent
RELEASED	Provider use known not to require charge	Stable key cannot create a second reservation
Unknown/recovery	Outcome needs operator/automatic reconciliation	No speculative release and re-charge

Implemented path
The client registers payment BFF routes and calls Payment Gateway. Fixture purchase, reservation, commit/release, operation keys, expiry and recovery are composed.

Current manual profile
real-providers deliberately keeps Stripe in fixture mode. Live checkout and real charges were not exercised by the evidence in this dossier.

Claim boundary. This proves wallet and provider-boundary engineering. It does not prove a production billing launch, reconciliation operations at scale or successful live Stripe checkout.

Persistable place truth is separated from transient provider data



Canonical location

Stable label, country, coordinates/postcode authority and provider provenance suitable for profile persistence.

Autocomplete session

Bounded provider token/session and suggestions remain transient; attribution is required.

Commute matrix

At most five eligible destinations, selected travel modes and advisory route estimates—never suitability scores.

Location and commute

Edit

Reading · within miles

Town or postcode

Reading

5 matching locations found.

Suggestions provided by Google Maps

Reading

Reading Street, Tenterden

Whitley, Reading

Reading Street, Broadstairs

Newtown, Reading

Live Google evidence

Five attributed suggestions

A suggestion is resolved to a canonical place before it can become persisted profile data. The provider attribution remains visible at the point of selection.

Why another service? The original Location Gateway directly represented Postcodes.io-style operations. Adding Google Places/Routes required a provider-neutral orchestration layer so the browser contract, canonical persistence and provider-specific credentials/DTOs stayed separate.

Reporting observes domain truth; it does not own or mutate it

Your job search progress

29 hours remaining this week.

Refresh

Applications

6

Generated documents

6

100%

Applied

0

0%

Interviews

0

0%

Offers

0

0%

Unsuccessful

0

0%

Recent activity

11/08 20:36

Documents

Downloaded previous cover letter version 1.

Source: DOCUMENT_STORE

Status: PREVIOUS_VERSION

11/08 20:36

Documents

Downloaded previous CV version 1.

Source: DOCUMENT_STORE

Status: PREVIOUS_VERSION

11/08 20:35

Targeted view: application totals, status buckets and complete recent-activity items are projected from owner services.

Sources

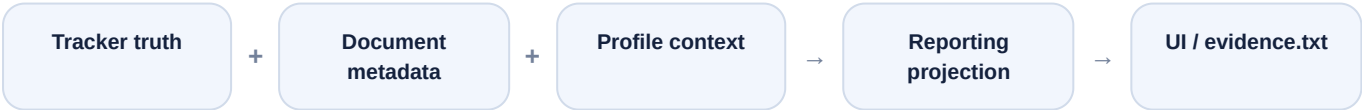
Tracker applications/events, Store document metadata and Profile data through owner-scoped service APIs.

Read models

Application totals/status/date buckets, recent content-free activity, indicative time and text evidence export.

Privacy

Document activity reports version/type/lifecycle—not CV or letter content. Empty, small and rich histories have explicit behaviour.



Indicative means indicative. Time estimates are explanatory product aids, not payroll, benefits or regulatory calculations.

Evidence basis: Reporting OpenAPI 2.1, summary/journal/evidence routes, document-activity projection and the retained reporting journey.

Deterministic personas are engineering infrastructure

What it provides

- ✓ versioned synthetic candidate personas;
- ✓ provider responses and error states;
- ✓ named reset/seed/verify lifecycle;
- ✓ service-owned cleanup orchestration;
- ✓ stable model fixture responses;
- ✓ guarded non-production APIs.

What it cannot prove

- current provider uptime or changing advert shape;
- real model quality, latency or cost;
- rich real-person evidence distribution;
- production security/operations;
- employment outcomes.

Unsafe testing risk

Captured personal/provider data, accidental live calls, paid model spend and non-repeatable failures.



Governed test state

Synthetic datasets, explicit provider modes, authenticated owner reset and deterministic verification.

Why canonical personas remain

Minimal, typical, rich, stress, uploaded-CV, manual-first and career-changer profiles exercise different structures while staying source-controlled and non-personal. A persistent manual account is separate runtime data and is never a fixture dependency.

7

canonical persona shapes

0

paid provider calls in fixture regression

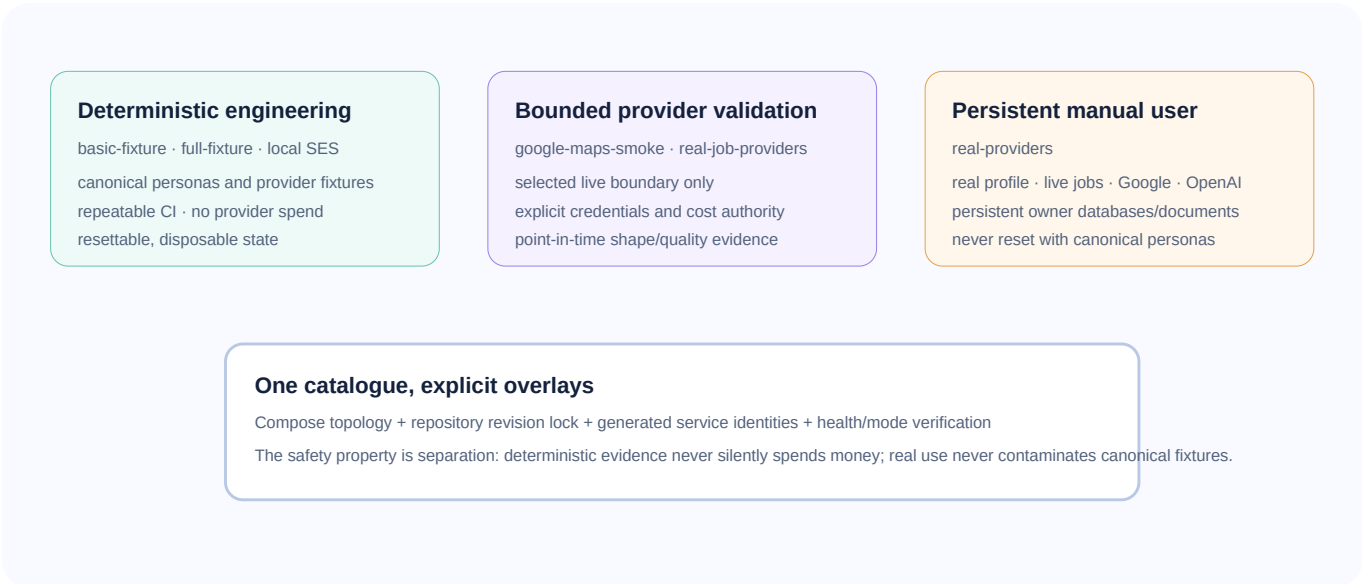
1

named-state owner for orchestration

strict

non-production safety boundary

Separate confidence questions require separate modes



Profile	Purpose	External state
basic-fixture	Account/profile/location/search fast path	Deterministic
full-fixture	Complete regression/product showcase	Deterministic
full-local-ses	Production email adapter against LocalStack	Local deterministic
google-maps-smoke	Only Google live	Explicit bounded live
real-job-providers	Five jobs + Postcodes live; LLM/Stripe fixture	Explicit live jobs
real-providers	Live jobs, Postcodes, Google and OpenAI; Stripe fixture	Persistent authorised manual user

Audited evidence. The retained manual-environment check reported all 28 application services healthy. Its PostgreSQL and document volumes survive ordinary restarts; secrets and personal data remain outside Git.

The demonstration is a passing browser journey, not a screen montage

Scenario: Alex moves from a new account to accepting an offer
register → build a rich evidence library → discover a live vacancy
→ save the vacancy → generate and review a CV and cover letter
→ inspect documents and reporting → progress the application
→ record applied, interview, offer and acceptance states

PASS

scenario and preflight exit codes

8:36

1920 × 1080 master recording

7

chapter clips cut from one timeline

2

generated PDFs downloaded and shown

Readable specification

The current feature contains 22 top-level behaviour steps and remains the authoritative narrative for the recording.

Browser execution

Playwright drives the real Angular UI, BFF and downstream services, including downloads and durable state transitions.

Recorded evidence

Timeline markers derive onboarding, profile, discovery, generation, documents, reporting and tracking clips from the same passing run.

Feature

→

Preflight

→

Isolated runtime

→

Browser
assertions

→

Master + clips +
PDFs

Disclosure. The 12 August recording uses a synthetic persona to protect personal data, but its vacancy providers and OpenAI path are live. Its report records `fixtureMode: false`, `providerMode: live` and successful scenario exit. This is bounded live evidence, not a claim of production availability.

Each layer answers a different confidence question

Domain	Unit/component	Integration/contract	E2E/live evidence
Identity/profile	validation, revisions, cookies	PostgreSQL, JWKS, owner denial	register/profile/reload/return
Job discovery	adapters, mapping, dedupe	producer clients, partial failure	five-source fixture; bounded live samples
Applications	transitions, idempotency	Store reference verification	save, select, apply, interview, offer, accept
Documents	lifecycle, MIME, renderer	PostgreSQL/filesystem/ClamAV	upload/generate/history/download
Generated content	parser, repair, claims, fallback	wallet/store/tracker recovery	fixture matrix + bounded paid samples
Location	canonical mapping, session bounds	Postcodes/Google clients	fixture commute + live Google smoke
Reporting	buckets, activity, redaction	owner-scoped upstream clients	empty/rich history and UI
Infrastructure	catalogue/policy/render tests	clean build and health/mode checks	capacity and restart persistence

32/32

Cucumber scenarios at audited 11 August revisions

Proves quickly

Validation, mapping and deterministic edge behaviour.

244/244

journey steps in that retained run

Proves together

Schema compatibility, persistence and multi-service ownership.

460/460

Angular tests in closure evidence

Proves realistically

User comprehension, live shapes, rich data and dated workload behaviour.

248/248

CV/letter domain tests in closure evidence

Revision boundary. At the audited 11 August closure revisions the deterministic regression passed 32 scenarios and 244 steps, alongside component, contract, persistence, security and accessibility suites. Several repositories changed on 12 August, so those totals are not presented as a current-HEAD rerun.

Changing data and paid integrations are tested without contaminating regression

Deterministic tests answer “does the same behaviour still work?” A persistent manual account and explicitly authorised live boundaries answer a different question: “does the product remain useful with a rich profile, current adverts and non-deterministic providers?”

Boundary exercised	Observed evidence	What it does not prove
Persistent account/profile	Structured work, project, education and skills data retained through ordinary service restarts	Production backup or multi-user operations
Current vacancies	Provider-backed discovery and detail through owned Reed, Adzuna and JSearch gateways	Provider uptime, completeness or long-term contract stability
Location	Google-attributed autocomplete and canonical resolution through Location Service	Every route, country or travel-mode combination
Document workflow	Realistic rich evidence exposed selection, deadline, recovery, export and association defects	Universal document quality or employment outcome
Bounded live showcase	Synthetic identity + live providers/OpenAI completed search → generation → documents → tracking	Public availability, load or production support

Useful separation

Real personal information stays in runtime databases and documents, never canonical personas, screenshots or Git.

High-value failures

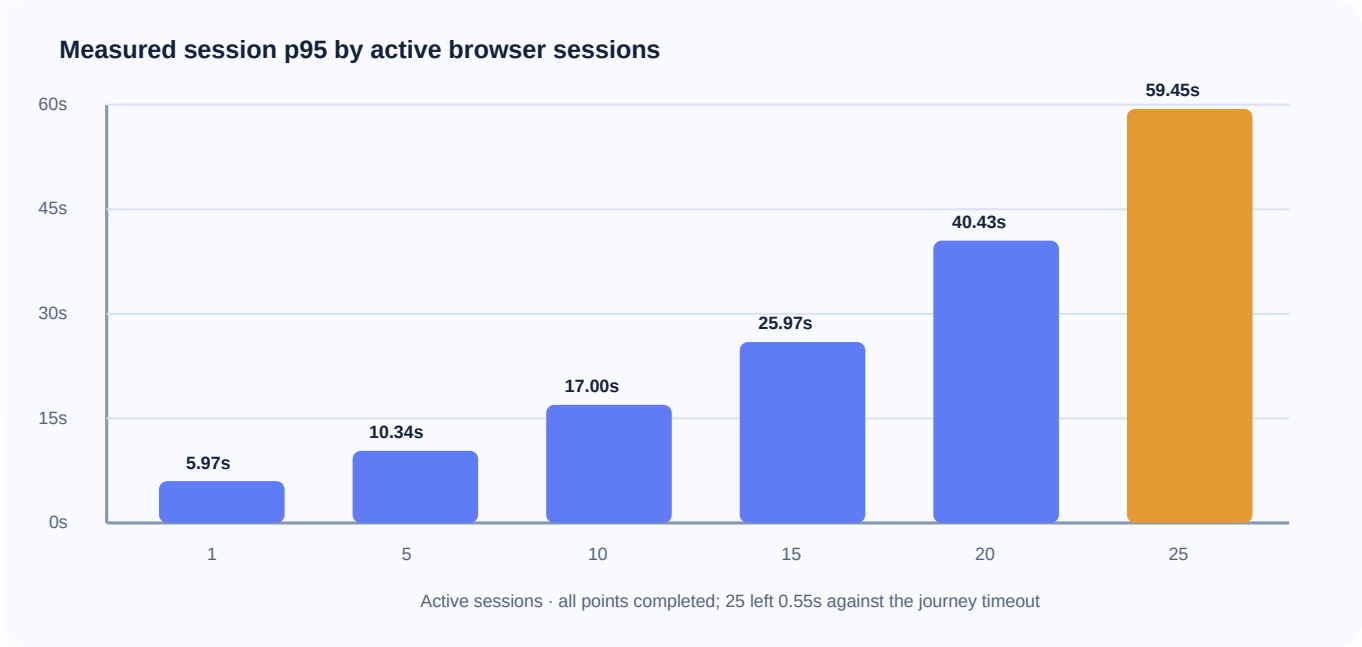
Rich data exposed issues that compact fixtures missed; corrections became general recovery and regression behaviour.

Honest confidence

Manual evidence broadens confidence, while dated reports and explicit modes prevent it being mistaken for repeatable CI proof.

Confidence gained. The two evidence systems complement each other: deterministic personas protect repeatability; realistic data and live boundaries challenge assumptions about usefulness and integration shape.

A failed benchmark found a product defect before it found a server limit



Sessions	Completed	Session p95	Response p95	Peak Docker memory
1	1/1	5.97 s	182 ms	8.01 GiB
5	5/5	10.34 s	571 ms	8.14 GiB
10	10/10	17.00 s	709 ms	8.35 GiB
15	15/15	25.97 s	472 ms	8.42 GiB
20	20/20	40.43 s	609 ms	8.66 GiB
25	25/25	59.45 s	546 ms	8.64 GiB

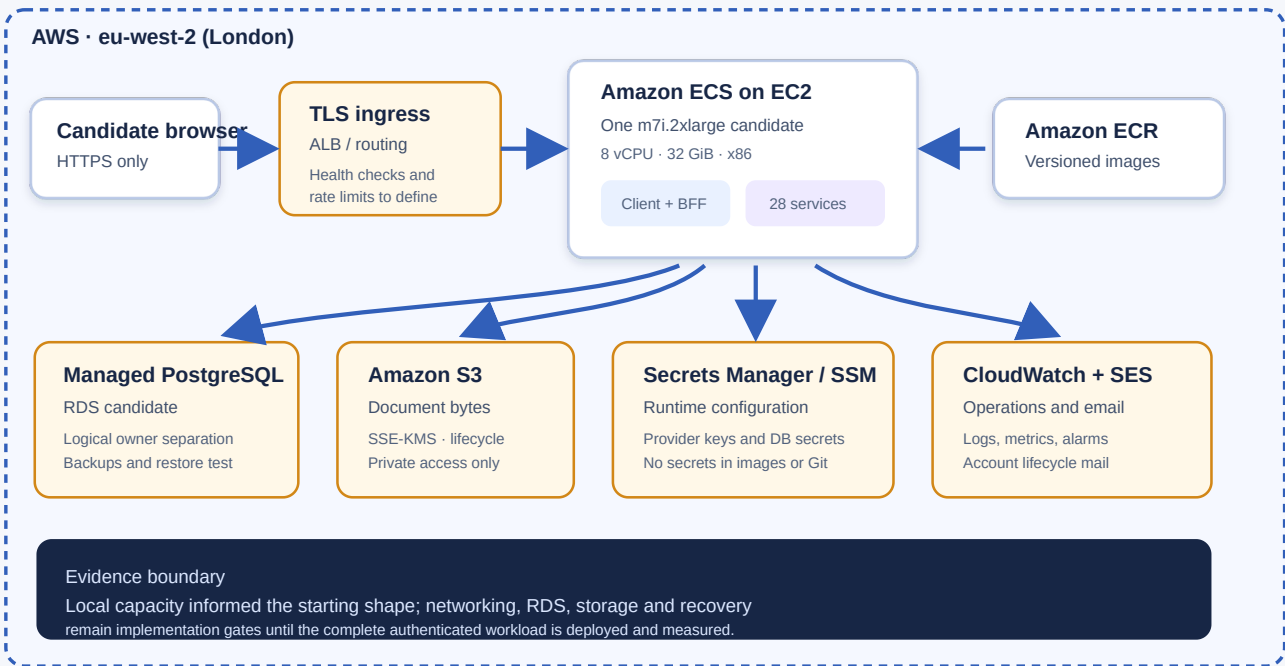
0/25 → 25/25. The original failure was not an application-service crash. A refreshed result gained canonical identity, causing Angular to recreate the job card during interaction. Stable provider-job tracking fixed the browser race.

Interpretation. This was one fixture-backed iteration per level, not a soak test. At 25, all workload scenarios completed with zero application errors, but session p95 left 0.55 seconds against the 60-second deadline. The report also retains expected bootstrap/external-resource response errors and one Docker-stats timeout. Browser workers were outside Docker measurements.

Local evidence informs a candidate; it does not validate the cloud

Candidate AWS private-beta implementation

Projected architecture · no main-application AWS workload has yet been deployed or load-tested



Application compute

Candidate: ECS on one London-region x86 EC2 m7i.2xlarge. ECR supplies versioned images; TLS ingress and health checks sit in front.

Owned data

Move seven logical PostgreSQL owners to managed database boundaries and document bytes to private S3 with managed encryption and lifecycle.

Operations

Secrets Manager/Parameter Store, CloudWatch logs/metrics/alarms, SES account email and tested backup/restore complete the intended control plane.

AWS capability already demonstrated at the public edge. The separate landing/waitlist system has deployed in eu-west-2: Angular → API Gateway → Lambda → DynamoDB/SES, with CloudFormation update, contact delivery and controlled SES bounce/complaint handling evidenced. It is intentionally not presented as deployment of the authenticated platform.

Status: planned. The authenticated application has not been deployed or load-tested on AWS. No main-application AWS resource was created for this evidence and validation spend was \$0. The separately deployed landing/waitlist stack is not evidence that the application platform is hosted.

A credible starting shape with explicit work before public use

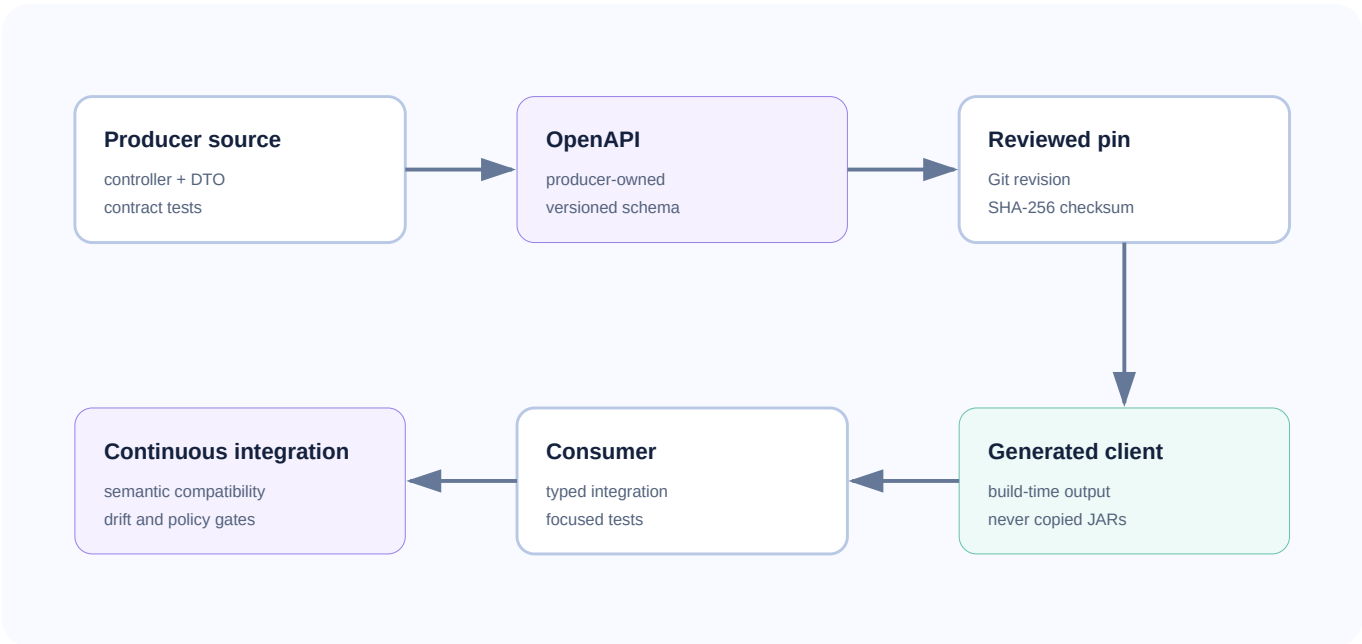
8	32 GiB	≈15	\$340.33
vCPU in the starting compute candidate	memory in one m7i.2xlarge	modelled initial active discovery journeys	dated 730-hour compute-only arithmetic
Why this starting point ✓ Local Docker memory reached 8.64 GiB, while service CPU peaked near eight logical cores. ✓ The 25-session run completed but had no useful deadline margin. ✓ An initial cap around 15 is intentionally below the functional ceiling. ✓ ECS on EC2 accommodates the current multi-container topology while reservations are refined.		What the figure excludes • RDS, EBS/S3 and backup storage; • load balancing, public IPv4, NAT and transfer; • DNS, logs, monitoring, support and tax; • high availability or a second node.	

Required implementation and validation sequence

1 Define task reservations, ingress, private networking and database ownership.	2 Build digest-pinned images and inject runtime secrets without placing them in Git or images.
3 Provision bounded eu-west-2 infrastructure with budget alarms and destroy controls.	4 Validate migrations, object storage, email, health checks, backups and restore.
5 Measure 10/15/20 journeys, failure behaviour and recovery before revising the cap.	

Decision discipline. The candidate is useful because it turns local measurement into an implementable hypothesis. It becomes an AWS capacity claim only after the intended network, compute, database and storage shape is deployed and measured.

Independent repositories require explicit compatibility discipline



Producer ownership
Controller/DTO source and contract tests produce the reviewed OpenAPI. Consumers do not edit private copies.

Revision + checksum
Consumer metadata names the producer Git revision and schema SHA-256, making drift reviewable.

Generated clients
Maven/TypeScript clients are disposable build output. Checked-in local JAR coupling was removed.

Delivery engineering

Concern	Practice	Failure prevented
Branches	Feature branch → PR → green CI → develop; main separately protected	Unreviewed integration changes
Parity	Workspace lock and catalogue coordinate reviewed revisions	"Works only in my checkout"
Semantic compatibility	Policy tests inspect required routes/schemas/enums	Compiles but changes meaning
Security/build gates	Secret history, dependency, licence, clean clone/container checks	False-green or irreproducible releases

Architecture consequence. Contract mismatch bugs—claim ledger bounds, timestamp offsets, application provenance and generated client metadata—became explicit PRs and regression gates rather than runtime surprises.

Failures that changed ownership or workflow

- 1 Generated document not reliably associated with its application.**
Found by realistic E2E. Generation linked selections before authoritative application identity/reconciliation was ready. The flow now creates/relinks the application and verifies exact references before selection; fixture and E2E preserve regression coverage.
- 2 Capacity refresh destroyed the job card under interaction.**
Concurrent Playwright journeys exposed unstable Angular tracking when canonical identity appeared after refresh. Stable provider-job identity changed the same workload from 0/25 to 25/25.
- 3 Google location did not fit the direct postcode design.**
Provider-specific concepts risked leaking into persisted profiles. Location Service was extracted so canonical place/route semantics remain independent of Google and Postcodes.io.
- 4 Compact fixture profiles hid realistic selection pressure.**
A rich profile exposed omitted and poorly prioritised evidence. Purpose-specific snapshots and relevance selection now preserve traceable inputs without making real personal data a test dependency.
- 5 Selected-output recovery assumed both document types existed.**
CV-only and letter-only journeys exposed an old paired-output assumption. Operation state and UI now preserve truthful partial success and deliberate omission.
- 6 Provider dates, IDs and offsets did not share one meaning.**
Compatibility normalisation was added at provider/domain boundaries while source attribution and original references remain visible for detail and application history.

Each fix added regression protection at the lowest useful layer and, where user behaviour mattered, at browser/E2E level.

Failures that changed recovery, quality or boundaries

- 7 External-call deadlines did not align with durable operation deadlines.**
A longer provider allowance alone was insufficient. Gateway bounds, caller read deadlines and persisted absolute deadlines were aligned; ambiguous outcomes enter reconciliation before retry.
- 8 One harmless structural defect could discard a whole document operation.**
A narrowly scoped exact-duplicate repair was introduced, followed by the same structural and evidence validation. Semantic rewriting remains prohibited.
- 9 Professional export quality regressed under rich content.**
Visual review found weak hierarchy and pagination that unit assertions could not express. A4 flow, spacing and page-break controls were corrected and retained as renderer tests.
- 10 Contract/client drift recurred across independently versioned services.**
Producer OpenAPI, reviewed revisions and checksums, regenerated clients and semantic policy gates replaced copied binaries and undocumented assumptions.
- 11 Repository health could disagree with product readiness.**
A service returning 200 was not enough. Mode checks, named-state verification, persistent-user journeys and focused cross-owner assertions were added to distinguish liveness from useful behaviour.
- 12 A capability could look ready before its complete user path was ready.**
Payment backend/BFF wiring and live Stripe availability are now described separately from the unfinished purchasing interface and standard fixture-mode configuration.

Maturity signal. The important result is not that defects existed; it is that realistic evidence identified why earlier tests missed them and converted them into architecture and regression controls.

Implemented controls with explicit assurance limits

Identity

HttpOnly cookies, CSRF, verified subjects and route-scoped service tokens.

Lineage

Digests and stable IDs across evidence, jobs, claims, documents and applications.

Untrusted content

Provider text, uploads and model output cross validation/allowlist/scanner boundaries.

Personal data

Real profile/CV/contact data remain in owner databases and outside fixtures, logs and Git.

Secrets

Runtime-only environment mechanism; provider keys remain in owned gateways.

Wallet

Reservations, operation keys, idempotent transitions and unknown-outcome checks.

Implemented, disabled, limited and post-beta

Classification	Examples
● Implemented/demonstrated	Profile/evidence, five-source discovery, applications, document generation/export, reporting, Google autocomplete, persistent local state.
● Implemented but not exposed	Live-capable Stripe checkout/webhook boundary; browser purchasing remains disabled.
● Known beta limits	No semantic suitability score; bounded provider/model samples; manual accessibility and production security assurance incomplete.
● Operational work	AWS application deployment/workload validation, HA/backups/monitoring, larger concurrency and first-user operations.
● Post-beta product	CV-to-profile import, richer matching, submission confirmation and application-answer tooling.

No compliance claim. These controls demonstrate design intent and implemented boundaries—not certification, complete penetration testing, production HA or regulatory approval.

A working controlled beta with a defined route to public operation

What a reviewer can rely on

- ✓ the connected application and 28-process runtime exist;
- ✓ core candidate state has explicit service/database owners;
- ✓ deterministic regression, live boundaries and manual use are separated;
- ✓ realistic testing has produced general product corrections;
- ✓ capacity evidence and cloud assumptions are dated and qualified;
- ✓ the next deployment gates are concrete and testable.

What remains before public beta

- deploy and measure the authenticated AWS workload;
- define production networking, HA, backup/restore and observability;
- complete purchasing UI and live billing assurance if enabled;
- broaden accessibility and independent security assurance;
- operate with pilot users and measure support and employment usefulness.

Confidence chain



The appropriate claim is not “production complete”. It is that a substantial product has been built, its important workflows have been exercised across real boundaries, its limits are known, and the remaining route to a controlled public beta is explicit.

Pages 39–45 make the system inspectable without private GitHub access: every repository has a purpose, caller, dependency and state description.

Client, identity and profile

The 31 repositories comprise 28 composed main-stack runtimes, one separately deployed public landing runtime, and two engineering repositories. Each entry names purpose, communications and state.

Browser boundary

job-seeker-copilot-client

Purpose. Authenticated Angular product and same-origin Express BFF.

Called by. Candidate browser.

Calls. User Management, Job Finder, Location, Document Generation, Document Store, Reporting and Payment gateways/services.

State. No application database; HttpOnly session cookies and transient UI state.

Identity owner

authentication-service

Purpose. Accounts, credentials, refresh/session lifecycle, reset and RS256/JWKS identity.

Called by. User Management; resource services read JWKS; lifecycle tooling.

Calls. Profile, Tracker and Store during account lifecycle; optional AWS SES.

State. PostgreSQL/Flyway owns accounts, credentials and session records.

Identity facade

user-management-gateway

Purpose. Browser-facing registration, session, profile and evidence facade.

Called by. Client BFF, including payment owner resolution.

Calls. Authentication Service and User Profile Service.

State. Stateless; derives trusted identity and returns stable, redacted failures.

Profile owner

user-profile-service

Purpose. Progressive profile, preferences, Evidence Library, immutable revisions and purpose-bound snapshots.

Called by. User Management, Job Finder, Document Generation and Reporting.

Calls. Authentication JWKS for resource identity.

State. PostgreSQL/Flyway owns profile, evidence revisions and snapshot digests.

Trust boundary. The browser never selects arbitrary service destinations or stores bearer tokens. Identity is derived at the BFF/facade and verified again at protected resource services.

Job discovery and matching

Job facade

job-finder-gateway

Purpose. Authenticated search, saved-job and application facade.

Called by. Client BFF.

Calls. Authentication JWKS, Job Service, User Profile and Application Tracker.

State. Stateless; retains provider and partial-failure truth in responses.

Job owner

job-service

Purpose. Five-provider fan-out, mapping, dedupe, detail hydration and immutable saved vacancies.

Called by. Job Finder and Document Generation.

Calls. Five provider gateways, Job Matching and Authentication JWKS.

State. PostgreSQL/Flyway owns saved jobs; bounded search/detail cache.

Enrichment

job-matching-service

Purpose. Reconciles results with application state and adds advisory commute assessments.

Called by. Job Service.

Calls. Application Tracker and Location Service.

State. Stateless; it does not currently produce a semantic suitability score.

Provider boundary

reed-gateway

Purpose. Reed search/detail authentication, mapping and source identity.

Called by. Job Service.

Calls. Reed API in live mode or System Data in fixture mode.

State. Stateless; bounded timeouts/errors preserve provider attribution.

Provider boundary

adzuna-gateway

Purpose. Adzuna credentialed paging and canonical response mapping.

Called by. Job Service.

Calls. Adzuna API in live mode or System Data in fixture mode.

State. Stateless; workplace type and source provenance are retained.

Specialist providers and location orchestration

Provider boundary

jsearch-gateway

Purpose. OpenWeb Ninja JSearch authentication and mapping.

Called by. Job Service.

Calls. OpenWeb Ninja JSearch in live mode or System Data in fixture mode.

State. Stateless; normalises aggregated IDs, dates and descriptions.

Specialist provider

nhs-jobs-gateway

Purpose. Maps official NHS Jobs Self-Serve XML adverts.

Called by. Job Service.

Calls. NHS Jobs API or an in-service deterministic fixture.

State. Stateless; authoritative source identity flows into saved jobs.

Specialist provider

apprenticeships-gateway

Purpose. Acquires and searches DfE apprenticeship adverts.

Called by. Job Service.

Calls. DfE Display Advert API v2 or an in-service fixture.

State. Atomic bounded in-memory vacancy snapshot; no user state.

Location facade

location-gateway

Purpose. Browser-facing legacy-compatible and v2 UK location routes.

Called by. Client BFF.

Calls. Location Service for both current route families.

State. Bounded postcode cache/rate-limit state; no persisted profile ownership.

Location domain

location-service

Purpose. Provider-neutral autocomplete, canonical resolution and commute orchestration.

Called by. Location Gateway and Job Matching.

Calls. Postcode.io Gateway and Google Maps Gateway.

State. Transient, bounded suggestion sessions; canonical results remain provider-neutral.

Location providers, applications and orchestration

Location provider

postcode-io-gateway

Purpose. Canonical UK place and postcode-provider boundary.

Called by. Location Service.

Calls. Postcodes.io in live mode or System Data in fixture mode.

State. Bounded positive/negative cache; provider credentials and DTOs remain isolated.

Location provider

google-maps-gateway

Purpose. Sole Google Places/Routes credential, DTO and attribution boundary.

Called by. Location Service.

Calls. Google Places and Routes APIs when explicitly enabled.

State. Transient, bounded provider-session tokens; no persistent candidate data.

Application owner

application-tracker-service

Purpose. Owns saved applications, lifecycle transitions, immutable events and exact document selections.

Called by. Job Finder, Matching, Document Generation, Reporting and account lifecycle.

Calls. Document Store for independent reference verification; Authentication JWKS.

State. PostgreSQL/Flyway is the authoritative application record.

Workflow coordinator

document-generation-gateway

Purpose. Durable generation, approval, upload, export, linking and recovery coordinator.

Called by. Client BFF.

Calls. Authentication, Profile, Job, CV, Payment, Store, Export and Tracker.

State. PostgreSQL/Flyway operation ledger; document bytes remain in Store.

Ownership rule. The coordinator persists intent and checkpoints, but it does not become the owner of profile evidence, vacancies, wallet balances, document files or applications.

Document generation, storage, export and reporting facade

Document content

cv-cover-letter-service

Purpose. Projects evidence, requests structured content, validates claims and supplies a conservative fallback.

Called by. Document Generation Gateway.

Calls. LLM Gateway; legacy/domain paths also call Payment, Store and Tracker.

State. No primary DB; optional encrypted short-lived rejected-output quarantine.

Model provider

llm-gateway

Purpose. Typed provider-neutral model transport, admission limits, usage and cost evidence.

Called by. CV/Cover Letter Service.

Calls. OpenAI in live mode or System Data in fixture mode.

State. Transient admission/circuit state; no candidate-profile ownership.

Document owner

document-store-service

Purpose. Owns document families, immutable versions, lineage, files and retention.

Called by. Client, Generation, Export, Tracker, Reporting and lifecycle paths.

Calls. Authentication JWKS, Application Tracker workflows and ClamAV.

State. PostgreSQL/Flyway plus local filesystem or S3-compatible object storage.

Artifact renderer

document-export-service

Purpose. Renders structured CV/letter content to DOCX/PDF and ingests bounded DOCX.

Called by. Document Generation Gateway.

Calls. Document Store for exact source/output versions.

State. Stateless; Apache POI/OpenPDF with replay-safe operation keys.

Reporting facade

reporting-gateway

Purpose. Authenticated browser facade for summary, journal and evidence-text export.

Called by. Client BFF.

Calls. Authentication JWKS and Reporting Service.

State. Stateless; it does not own reporting data.

Reporting, wallet, payments and deterministic data

Read model

reporting-service

Purpose. Builds read-only application, profile and document summaries.

Called by. Reporting Gateway.

Calls. Application Tracker, Document Store and User Profile.

State. No authoritative or cached persistence; upstream owners remain truth.

Wallet owner

payment-service

Purpose. Owns credit wallets, append-only ledger, reservations and reconciliation.

Called by. Payment Gateway, Stripe, Document Generation, CV and tooling.

Calls. No external provider.

State. PostgreSQL/Flyway owns balances and idempotent accounting transitions.

Payment facade

payment-gateway

Purpose. Wallet, pricing and checkout facade.

Called by. Current Client BFF payment routes.

Calls. Payment Service and Stripe Gateway.

State. Stateless; ordinary purchasing UI and standard live Stripe mode remain unavailable.

Payment provider

stripe-gateway

Purpose. Isolates Stripe checkout/session and signed webhooks.

Called by. Payment Gateway and Stripe webhooks.

Calls. Stripe live or System Data fixture; Payment Service fulfilment.

State. Stateless; standard manual profile uses fixture mode.

Non-production data

system-data-service

Purpose. Versioned synthetic personas, provider fixtures and named-state orchestration.

Called by. Fixture-mode gateways, E2E and authorised operators.

Calls. Guarded reset/seed/verify APIs for Auth, Profile, Tracker, Store and Payment.

State. Source-controlled synthetic datasets and bounded indexes; never production truth.

Verification, runtime engineering and public landing

Cross-service verification

e2e

Purpose. Browser journeys, accessibility, workload and executable showcase automation.

Called by. CI and engineering operators.

Calls. Product browser routes and guarded System Data lifecycle APIs.

State. Disposable reports, traces, recordings and downloads only.

Runtime and delivery

infrastructure

Purpose. Fleet catalogue, Compose profiles, revision locks, builds, operations and documentation.

Called by. Developers, CI and operators.

Calls. Coordinates all runtime repositories plus PostgreSQL, ClamAV and optional tooling.

State. Configuration and persistent volumes; no user-facing domain API.

Separate public runtime

job-seeker-copilot-landing

Purpose. Public product site, double-opt-in waitlist and contact journey.

Called by. Public browser and SES delivery events.

Calls. AWS Amplify/API Gateway/Lambda, DynamoDB, SES and EventBridge/CloudWatch.

State. DynamoDB workflow state; no dependency on the authenticated application.

Runtime support, not repositories

Seven PostgreSQL containers, ClamAV and optional LocalStack/Portainer/Dozzle are meaningful dependencies but excluded from the 31-repository count.

How to verify further

The dossier contains the ownership and communication information needed for first review. Private source, contracts and retained run reports can be reviewed by arrangement where deeper assurance is appropriate.

Confidence is built by connecting claims to explicit boundaries: what exists, what called what, which component owns the state, what was exercised, and what remains planned.

This dossier accompanies a job-specific CV, cover letter and executable product demonstration. Contact details remain in the accompanying application documents.